

JASON POULOS

Research at the intersection of machine learning, causal inference, and policy, with a current focus on evaluating frontier AI models.

postdoctoral training

Postdoctoral Fellow in Machine Learning, Division of Endocrinology, Brigham and Women's Hospital and Harvard Medical School, 2023 – 2024

Postdoctoral Fellow in Data Science, Department of Health Care Policy, Harvard Medical School, 2021 – 2023

Postdoctoral Associate, Causal Inference and Deep Learning Programs, Statistical and Applied Mathematical Sciences Institute, 2019 – 2021

education

Ph.D., Political Science with a Designated Emphasis in Computational Science and Engineering (NSF Graduate Fellow), University of California, Berkeley, 2019

M.A., Political Science, University of California, Berkeley, 2014

Graduate Special Student (non-degree graduate coursework in Economics), Massachusetts Institute of Technology, 2011–2013

Non-degree graduate coursework in Economics, Harvard University, 2010

B.A., Economics (Phi Beta Kappa), University of Massachusetts, Amherst, 2008

conference papers

Mike A. Merrill, Alexander G. Shaw, Nicholas Carlini, Boxuan Li, Harsh Raj, Ivan Bercovich, Lin Shi, Jeong Y. Shin, Thomas Walshe, Junhong Shen, Guanghao Ye, Haowei Lin, **Jason Poulos**, Maoyu Wang, E. K. Buchanan, Jenia Jitsev, Marianna Nezhurina, Di Lu, Orfeas M. Mastromichalakis, ..., Ludwig Schmidt (2026). “[Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces](#).” *International Conference on Learning Representations*. Contributed 8 benchmark tasks (see [Benchmark Contributions](#)). [[arXiv](#)] [[project](#)] [[blog](#)]

journal articles

Center for AI Safety, Scale AI, & HLE Contributors Consortium (2026). “[A benchmark of expert-level academic questions to assess AI capabilities](#).” *Nature*, 649: 1139–1146. Co-author as a member of the HLE Contributors Consortium. [[arXiv](#)] [[project](#)]

Denis Agniel, Sharon-Lise Normand, John Newcomer, Katya Zelevinsky, **Jason Poulos**, Jeannette Tsuei, and Marcela Horvitz-Lennon (2024). “[Revisiting Diabetes Risk of Olanzapine versus Aripiprazole for Serious Mental Illness Care](#).” *BJPsych Open*, 10(5): e144.

- Jason Poulos (2024). “State-Building through Public Land Disposal? An Application of Matrix Completion for Counterfactual Prediction.” *Statistics and Public Policy*, 11(1). [arXiv] [code]
- Jason Poulos, Marcela Horvitz-Lennon, Katya Zelevinsky, Thomas Huijskens, Pooja Tyagi, Jiaju Yan, Jordi Diaz, Tudor Cristea-Platon, and Sharon-Lise Normand (2024). “Targeted Learning in Observational Studies with Multi-Valued Treatments: An Evaluation of Antipsychotic Drug Treatment Safety.” *Statistics in Medicine*, 43(8): 1489–1508. [arXiv] [code]
- Jason Poulos, Sharon-Lise Normand, Katya Zelevinsky, John Newcomer, Denis Agniel, Haley Abing, and Marcela Horvitz-Lennon (2023). “Antipsychotics and the Risk of Diabetes and Death among Adults with Serious Mental Illnesses.” *Psychological Medicine*, 53(16): 7677–7684. [code]
- David Rios Insua, Roi Naveiro, Víctor Gallego, and Jason Poulos (2023). “Adversarial Machine Learning: Bayesian Perspectives.” *Journal of the American Statistical Association*, 118(543): 2195–2206. Awarded the *Premio SEIO-Fundación BBVA* (2026). [arXiv] [code]
- Zhenhua Wang, Olanrewaju Akande, Jason Poulos, and Fan Li (2022). “Are Deep Learning Models Superior for Missing Data Imputation in Surveys? Evidence from an Empirical Comparison.” *Survey Methodology*, 48(2): 375–399. [arXiv] [code]
- Jason Poulos and Shuxi Zeng (2021). “RNN-Based Counterfactual Prediction, with an Application to Homestead Policy and Public Schooling.” *Journal of the Royal Statistical Society, Series C*, 70(4): 1124–1139. [arXiv] [code]
- Jason Poulos and Rafael Valle (2021). “Character-Based Handwritten Text Transcription with Attention Networks.” *Neural Computing & Applications*, 33(16): 10563–10573. [arXiv] [code]
- Kellie Ottoboni and Jason Poulos (2020). “Estimating Population Average Treatment Effects from Experiments with Noncompliance.” *Journal of Causal Inference*, 8(1): 108–130. [arXiv] [code]
- Jason Poulos (2019). “Land Lotteries, Long-Term Wealth, and Political Selection.” *Public Choice*, 178: 217–230. [code]
- Jason Poulos and Rafael Valle (2018). “Missing Data Imputation for Supervised Learning.” *Applied Artificial Intelligence*, 32(2): 186–196. [arXiv] [code]

other
articles

Jason Poulos (2023). “Gender Gaps in Frontier Entrepreneurship? Evidence from 1901 Oklahoma Land Lottery Winners.” Special issue on “The Development of the American West,” *Journal of Historical Political Economy*, 2(4): 611–634. [arXiv] [code]

Jason Poulos (2021). “Amnesty Policy and Elite Persistence in the Postbellum South: Evidence from a Regression Discontinuity Design.” Special issue on “Slavery and its Legacies,” *Journal of Historical Political Economy*, 1(3): 353–375. [arXiv] [code]

Jason Poulos (2007). “Indexing the Massachusetts Minimum Wage.” Special issue on “25 Ideas for Working Families in America,” *Policy Studies Journal*, 35(3): 20–21.



Full list of articles on [Google Scholar](#).

benchmark
contributions

Terminal-Bench (Stanford University & Laude Institute), 2025 – 2026: 17 merged pull requests to the [benchmark repository](#), including 8 original tasks packaged as containerized RL environments with automated graders: [parallelize-graph](#) (parallel de Bruijn graph construction for genome assembly in Unified Parallel C; featured in the [task spotlight](#)); [fmri-encoding-r](#); [stable-parallel-kmeans](#); [parallel-particle-simulator](#); [adaptive-rejection-sampler](#); [cross-entropy-method](#); [attention-mil](#); [play-lord](#).

Humanity’s Last Exam (Center for AI Safety & Scale AI), 2025: contributed expert-level questions to a benchmark of frontier AI capabilities; credited co-author of the *Nature* (2026) paper. [project]

honors, grants
& fellowships

[Premio SEIO–Fundación BBVA](#) for Best Contribution from Statistics and Operations Research to Data Science, awarded for “Adversarial Machine Learning: Bayesian Perspectives” (with D. Rios Insua, R. Naveiro, and V. Gallego), 2026

National Science Foundation Graduate Research Fellowship, 2013–2018

NSF Frontera Startup Allocation: “RNN-Based Counterfactual Prediction on High-Dimensional Longitudinal Health Data” (SES20001), 2020–2021

NSF XSEDE Startup Allocation: “RNN-Based Counterfactual Time-Series Prediction” (SES180010), 2018–2019, 2020–2021

Berkeley Empirical Legal Studies Graduate Fellowship, University of California, Berkeley, School of Law, 2016–2017

professional
service

Conference Reviewer: AAAI Conference on Artificial Intelligence (AAAI; 2026); Artificial Intelligence and Statistics (AISTATS; 2022 – 2023, 2025 – 2026); IEEE International Conference on Data Science and Advanced Analytics (IEEE DSAA; 2025); International Conference on Machine Learning (ICML; 2025); International Conference on Learning Representations (ICLR; 2025 – 2026); Machine Learning for Health (ML4H; 2021 – 2024); Machine Learning for Healthcare (MLHC; 2025); Neural Information Processing Systems (NeurIPS; 2024, 2026; Datasets & Benchmarks Track, 2025); Uncertainty in Artificial Intelligence (UAI; 2021, 2024 – 2025)

Workshop Reviewer: ICML Workshop on Test-Time Adaptation: Putting Updates to the Test! (PUT; 2025); ICML Workshop on Technical AI Governance (TAIG; 2025); NeurIPS Workshop on Machine Learning and the Physical Sciences (2019, 2020)

Ethics Reviewer: NeurIPS Ethics Review (2023 – 2025)

Journal Reviewer: *American Journal of Political Science*; *Applied Artificial Intelligence*; *Applied Stochastic Models in Business and Industry*; *Bayesian Analysis*; *BMC Medical Research Methodology*; *Distributed and Parallel Databases*; *Economics & Politics*; *European Journal of Operational Research*; *Frontiers in Big Data – Data Mining and Management*; *GigaScience*; *Journal of Applied Econometrics*; *Journal of the Royal Statistical Society: Series C*; *PeerJ Computer Science*; *PLOS ONE*; *PLOS Neglected Tropical Diseases*; *Statistical Methods & Applications*; *Statistical Papers*; *Statistics and Public Policy*

Book Reviewer: Chapman & Hall/CRC Statistics; Springer Mathematics

invited talks

Department of Engineering & Public Policy, Carnegie Mellon University, March 2023

Dalla Lana School of Public Health, University of Toronto, February 2023

Brandeis International Business School, Brandeis University, December 2022

Modern Techniques in Survey Sampling, Canadian Statistical Sciences Institute, University of Ottawa, July 2022

Department of Mathematics, Université du Québec à Montréal, February 2022

Statistical Methods for Computational Advertising, Banff International Research Station, October 2021

conference talks

Causal Data Science Meeting (CDSM; 2021, 2022)

RAND Center for Causal Inference Symposium (2022)

Joint Statistical Meetings (JSM; 2021, 2022)

Political Institutions and Political Economy Collaborative, Bedrosian Center, University of Southern California (2021, 2022)

Society for Political Methodology (PolMeth; 2020, 2021; Europe, 2021, 2022; Asia, 2022)

Eastern North American Region International Biometric Society (ENAR; 2022)

Online Causal Inference Seminar (OCIS; 2021[†])

Big Data Meets Survey Science (BigSurv20; 2020)

Data Science, Statistics & Visualization (DSSV; 2020)

American Political Science Association (APSA; 2014^{*}, 2015, 2018[‡])

Midwest Political Science Association (MPSA; 2018)

^{}poster; [†]discussant; [‡]paper & discussant*

teaching & mentoring

Graduate Student Instructor: Department of Political Science, University of California, Berkeley: Intro. to American Politics (undergrad), spring 2017 and spring 2018; Intro. to Empirical Analysis & Quantitative Methods (undergrad), fall 2018

Research Mentor: Undergraduate Research Apprentice Program (URAP), University of California, Berkeley, fall 2016 and spring 2017

other professional experience

Human Frontier Collective Specialist (contract), Scale AI, March – December 2025: designed domain-specific problem sets to evaluate LLMs; trained machine learning engineer agents by creating solutions to machine learning benchmark competitions

Research Support Associate, Department of Political Science, Massachusetts Institute of Technology, August 2011 – May 2013: data programming and analysis (R, Stata) for a randomized controlled trial on information interventions in East Africa (E. Lieberman, D. Posner, L. Tsai), and for projects on citizen perceptions of government responsiveness (L. Tsai), the politics of redistributive spending (D. Posner), and economic liberalization and migrant remittances (D. Singer)

Research Consultant, Production in the Innovation Economy (PIE), Massachusetts Institute of Technology, December 2012: assembled dataset for a project on U.S. manufacturing (R. Wellhausen)

Research Consultant (remote), Center on the Demography and Economics of Health and Aging (CDEHA), Stanford University, April – June 2011: data programming for research on child care and development economics (S. Jayachandran)

Research Assistant, Department of Economics, Harvard University (based at the National Bureau of Economic Research), November 2010 – April 2011: data programming and analysis (Stata) for a project on the for-profit higher education sector (C. Goldin)

Research Assistant, Harvard Decision Science Laboratory, Harvard Kennedy School, October 2009 – November 2010: research design, subject recruitment, and experimentation for laboratory experiments in economics and behavioral decision making

Research Consultant (remote), Institute for Financial Research (SIFR), Stockholm, September – November 2010: ran behavioral experiments on Amazon Mechanical Turk and used command-line tools to distribute subject payments (A. Almenberg)

Economist Intern, Bureau of Labor Statistics, U.S. Department of Labor, Boston, March – September 2009: collected and coded firm-level survey data on worker injuries for the Occupational Safety and Health Statistics Program

technical skills

Languages: R; Python; Bash; C/C++; UPC; SQL

Version Control: Git (GitHub)

Frameworks & Libraries: PyTorch; TensorFlow; Keras; scikit-learn; Ray Tune; LiteLLM; Open MPI

Infrastructure: Docker; CUDA (multi-GPU); HPC (TACC Frontera, XSEDE); Harbor

Operating Systems: Linux (CentOS, Ubuntu)

certifications

Good Clinical Practice (ICH GCP E6(R2)) and Clinical Research Boot Camp (human subjects research), Mass General Brigham Human Research Affairs, 2023